

Silent-Failure Study

Status as of 22 August 2026

Where you are: the baseline is locked. Week 1 begins.

The measurement of your own engine is finished and reproducible. Everything remaining is about the other two engines.

Done

- ✓ **Engine frozen** at commit a8dc2b2, tag v2.1.3. Recorded in stress_suite/FREEZE.md.
- ✓ **Baseline measured and re-verified.** 55 cases: 44 pass, **0 silent**, 1 error, 10 timeout. 45 adjudicated.
- ✓ **Reproduction confirmed.** Re-run at the new pin returned bit-identical results to the 10 August run — zero status changes across 55 cases, no metric changed by any amount.
- ✓ **Third engine chosen:** Drake, over Pinocchio. Recorded in SCOPE.md.
- ✓ **Documentation:** baseline freeze record (13 pp) and engine/harness fixes writeup (5 pp), both printed.

Next — Week 1: adapters

This is the hard part and where all the risk lives. Budget the whole week.

- **Install Drake in Ubuntu WSL.** Not native Windows. It is the heavier of the two candidates — do this in the first days, not on day four.
- **SymPy adapter.** Express the pendulum chain via sympy.physics.mechanics. Already installed.
- **Drake adapter.** Same chain, same knobs.
- **The gate:** all three engines agree on one easy case you can check by hand. Do not proceed past this until they do.

Then — Week 2: sweep all three, build the comparison table, write.

Rules in force until the study closes

- ! **The engine does not change.** Fixing MechanicsDSL mid-study is what makes the numbers meaningless. If it moves off a8dc2b2, the baseline is void and the sweep must be repeated.
- ! **The instrument does not change either.** Same reasoning. Improvements to the harness are recorded, not applied.
- ! **Quote numbers only from RESULTS.md.** The 28 July report describes a pre-fix engine and must never be cited as current.

Known limit on the headline

The independent ground-truth oracle covered **22 of the 45** adjudicated cases, all on the Lagrangian pathway. The other 23 — every Hamiltonian case (14), every constrained case (8), and one Lagrangian case where it did not apply — rest on energy, constraint, and structural checks alone.

State the result **by pathway**, not pooled. Extending the oracle to the Hamiltonian pathway is the first thing to do *after* the study closes.

At a glance

Pin	a8dc2b2 tag v2.1.3 (git checkout v2.1.3)
Headline	Zero silent failures in 45 adjudicated cases
Caveat	Independent oracle covers 22 of those 45
Timeout	180 s per case
Engines	MechanicsDSL ✓ SymPy □ Drake □
Paper	Short or workshop paper. <i>Not</i> a registered report.
Plan	stress_suite/SCOPE.md
Pin record	stress_suite/FREEZE.md
Numbers	stress_suite/RESULTS.md
Why they changed	stress_suite/FIXES.md
Repository	Desktop\mechanicsdsl-main

Disclose in the paper: you maintain one of the engines measured, and AI assistance was used. Both are permitted; concealing either is not.